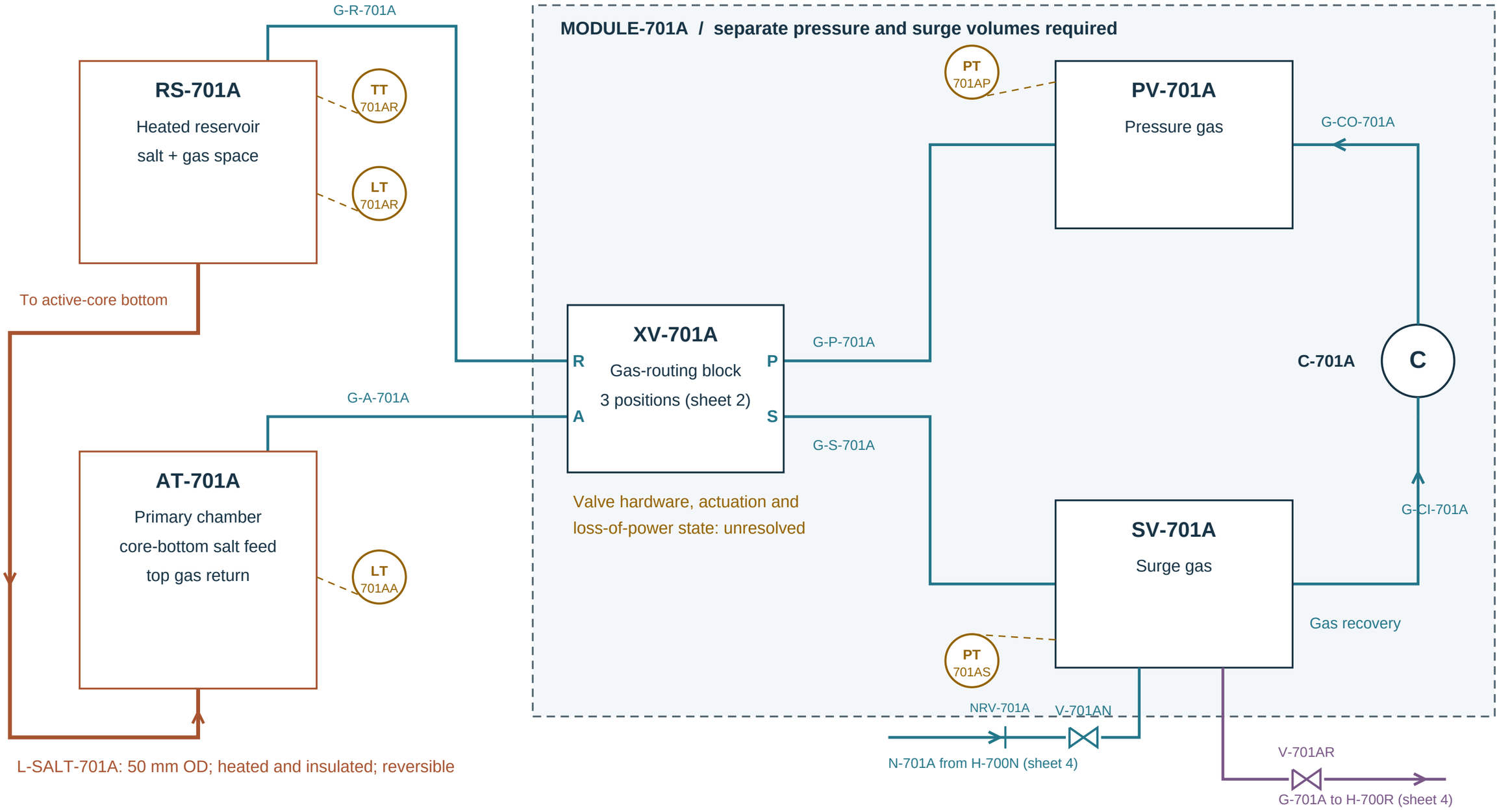
Absorber salt system

One circuit shown; repeat independently for primary A and secondary B at six radial assemblies



Arrows on salt line show injection; recovery reverses flow. Gas lines from reservoir/chamber: 25 mm OD visual allowance.
Amber instruments are proposals. Heater controls, relief paths and compressor nonreturn details remain open.
Charge and discharge service valves are closed in normal duty. Separate return goes to gas treatment before any release.
This sheet defines functions and endpoints. It does not assign shutdown logic, pressure ratings, valve sizing or insertion time.

Circuit repetition and operating functions

A = primary, B = secondary; radial centre/outer assignment remains to be confirmed

GAS ROUTING FUNCTION

Position	Connected ports	Intended effect
Remove	P-A and R-S	Pressure on absorber; salt returns to reservoir; reservoir gas enters surge vessel
Hold	All four blocked (proposed)	Isolated process gas; confirm hold behavior and loss-of-power response
Inject	P-R and A-S	Pressure on reservoir; displaced absorber gas enters surge vessel

TWELVE INDEPENDENT CIRCUITS

Assembly	Primary circuit	Secondary circuit
1	AT / RS / PV / SV / C / XV - 701A	AT / RS / PV / SV / C / XV - 701B
2	AT / RS / PV / SV / C / XV - 702A	AT / RS / PV / SV / C / XV - 702B
3	AT / RS / PV / SV / C / XV - 703A	AT / RS / PV / SV / C / XV - 703B
4	AT / RS / PV / SV / C / XV - 704A	AT / RS / PV / SV / C / XV - 704B
5	AT / RS / PV / SV / C / XV - 705A	AT / RS / PV / SV / C / XV - 705B
6	AT / RS / PV / SV / C / XV - 706A	AT / RS / PV / SV / C / XV - 706B

Each circuit has one absorber chamber, one salt reservoir, two gas volumes, one recovery compressor and a routing function.

Six stationary absorber tanks each contain two chambers. Each chamber has its own reservoir and absorber module.

Suggested monitoring: pressure in each gas vessel; reservoir temperature and inventory; chamber fill indication (method unresolved).

Gas recovery is from surge to pressure vessel. Compressor duty, reset sequence and isolation/check arrangements remain unselected.

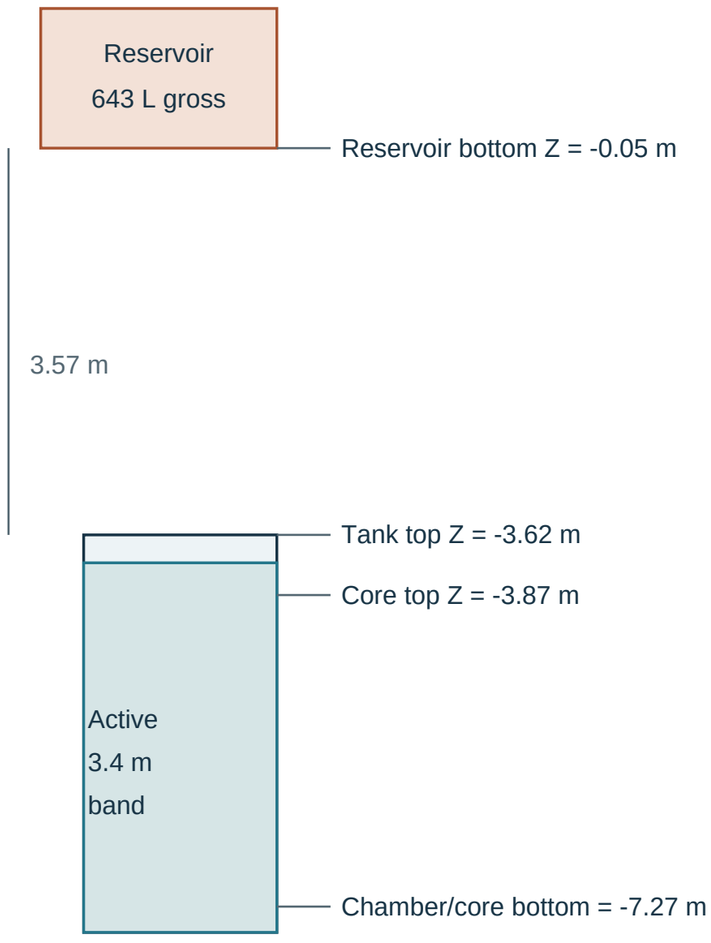
Normal duty: isolate both service lines per module. Charging and discharge functions and unresolved controls are on sheet 4.

Machine-readable equipment, line and instrument records expand all twelve circuits; a register accompanies this drawing.

Capacity check against the current 3D model

Source charges retained; measured site elevations; owner excludes the 0.30 m bottom-reflector region

VERTICAL GEOMETRY



VOLUMES PER ASSEMBLY / CIRCUIT

Item	Volume
Absorber tank (both chambers)	613.2 L gross
Active-height absorber tank envelope	571.2 L gross
Source primary + secondary	352.7 L salt
3 mm wall/partition trial	526.1 L active capacity
Each external salt reservoir	643.4 L gross
Each module lower cylinder	643.4 L gross

IDEAL GAS SPACE PER MODULE

Case	Pressure	Surge	Total
Primary active	158.9 L	356.1 L	515.0 L
Secondary active	193.8 L	422.2 L	616.0 L
Primary + gas	158.9 L	363.1 L	522.0 L
Secondary + gas	193.8 L	430.1 L	623.9 L

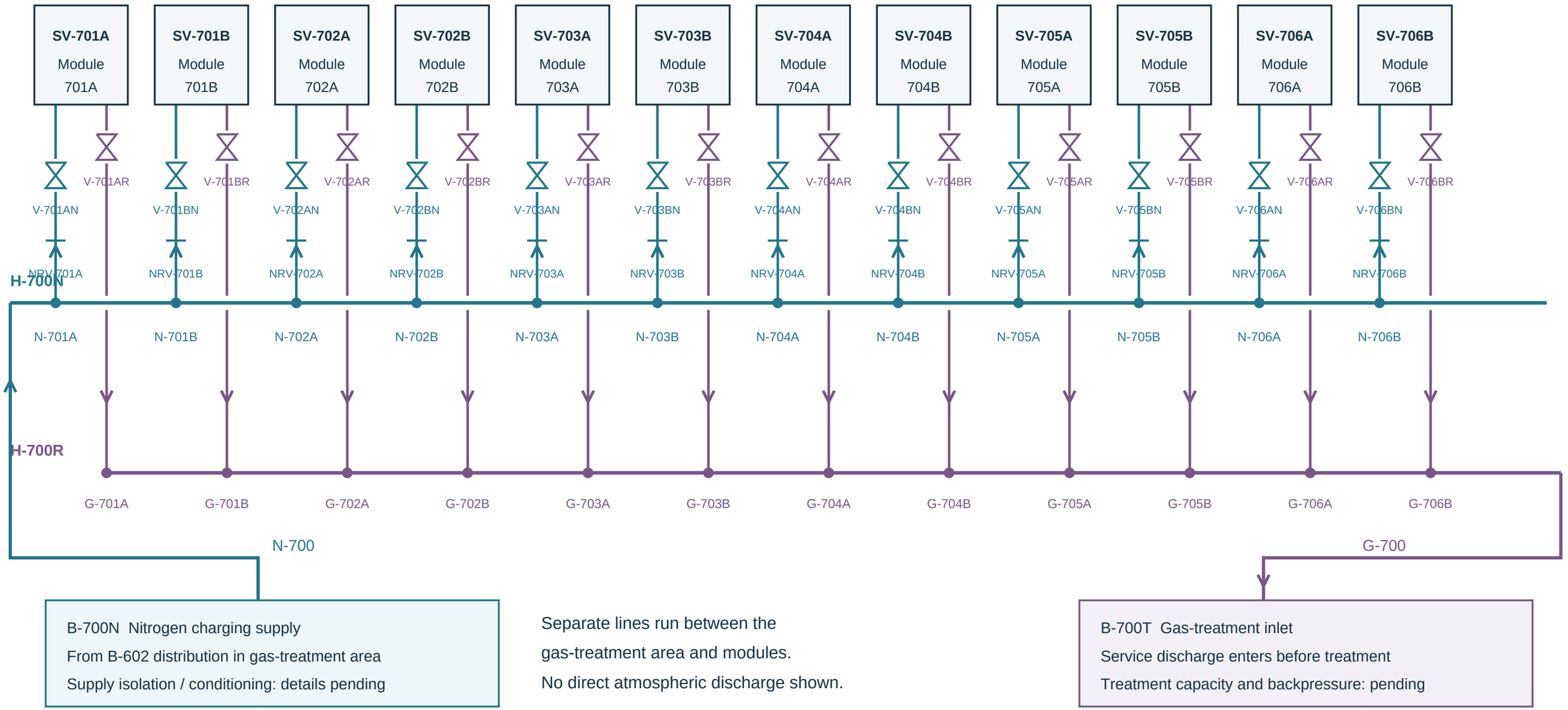
+ gas = 0.25 m upper space retained as gas, not salt.
Lower-cylinder fit is tight for secondary; upper box space may be needed. Heads, walls and equipment are additional.

Gas-space estimates use equal-temperature isothermal balances, source-derived density, zero heel and prefilled salt lines.
Both cases exclude the bottom reflector. The second includes provisional upper gas space; neither adds salt below the core.
No updated injection time: viscosity, final hydraulic route, gas restrictions and required shutdown time remain unresolved.
Source: AbsorberSaltInjectionTime.docx, pp. 1-6. See the editable Hurmet worksheet for equations, assumptions and checks.
The existing 3D salt tubes stop at the absorber tank top. External connections at the active-core bottom are shown on sheet 1, pending 3D routing.

Separate nitrogen charging and gas discharge

Twelve modules; separate clean supply and treatment-return headers; internal surge-vessel connections are provisional

SV service ports shown; one module per absorber chamber. All sizes and charging pressures remain unresolved.



B-700N Nitrogen charging supply
From B-602 distribution in gas-treatment area
Supply isolation / conditioning: details pending

Separate lines run between the gas-treatment area and modules.
No direct atmospheric discharge shown.

B-700T Gas-treatment inlet
Service discharge enters before treatment
Treatment capacity and backpressure: pending

Normal duty: both module service isolation valves closed. Charge: N open / R closed. Discharge: N closed / R open.

These are intended connection states, not procedures. Valve actuation, fail positions, sequencing and pressure limits are unresolved.

NRV symbols reserve charging-branch nonreturn functions; device selection and backflow-protection design remain open.

Surge-vessel service ports are provisional. Servicing other isolated gas spaces and module-side dead-leg volumes need review.